Data Architect

Cloud Adoption Framework / Azure Cloud / Data Lake / DataOps

EXPERTISE

1. Azure Architecture
2. CI/CD Strategy with Azure DevOps
3. Design and implementation of Data Platform solutions on Azure
4. Data Lake and Data Warehouse design and Snowflake solution implementation
5. Security and Compliance in Azure
6. Monitoring and Performance Management
7. Teamwork and User Training
8. Dashboarding with Power BI for Operational Infrastructure & Data Monitoring
9. Problem Solving and Technical Support

PROFESSIONAL EXPERIENCE

JUNE 2024 - PRESENT | BOMBARDIER, MONTREAL, CANADA

Data Architect

Project: Azure Platform Administration, Data Engineering and Data Modeling

- Implementation of CI/CD pipelines via Azure DevOps YAML pipelines for infrastructure deployment
- Design, build, test and deploy Data & BI solutions used for various business sectors and applications in ongoing and future projects
- Design, develop, deploy and optimize data and BI pipelines used to feed data applications or data visualizations
- Ensure execution and maintenance of Data & BI solutions and Data & BI products
- Design, develop and deploy visualization and analysis tools, as well as automation, to facilitate DevOps on Azure Data Platform
- Share best practices in Data & BI processing and data storage with other data engineers
- Design and develop data integration and consumption solutions using APIs, ETL jobs and/or streaming (ETL pipelines with Python / Spark / Azure Data Factory)
- Develop and maintain scalable data pipelines and create new integration APIs to support continuous growth in data sources, volume and complexity
- Model and normalize data to meet team and business line needs

- Diagnose problems and solve complex issues ensuring developed solutions are sustainable with a continuous improvement perspective
- Implement continuous deployment strategies while ensuring high availability of production systems
- Participate in architecture choices for implementing analytical projects under Power BI, including choice of Power BI license types (Free, PRO and Premium)
- Develop Power BI Dashboards exposing customer-related indicators
- Calculate and implement analysis and monitoring strategy for customer KPIs using tabular cubes and Power BI
- Administer Power BI service: pipelines, deployments, workspaces, security, Gateway, Premium Capacity

Azure Environment and Services: Snowflake, Azure VM, Azure VNet, PyCharm, Python, REST/RESTful, NoSQL, HDFS, Apache Spark, SQL Server, Apache Kafka, Agile, Scrum, Jira, Git, Azure DevOps, Linux, Windows, Docker, Azure: Azure Synapse, Azure Data Factory, Azure Data Lake Store Gen2, Azure Function, Databricks, Delta, Power BI

JUNE 2023 - FEBRUARY 2024 | SAINT-GOBAIN, FRANCE

Data Architect

Saint-Gobain Cloud Migration Project: Deployment of a data management platform integrating Azure, deployment automation and cost optimization with Azure services

Cloud Architecture Planning and Design

- Organization and structuring of Cloud platform implementation work using Microsoft Cloud Adoption Framework best practices
- Definition of Azure hierarchical structure including Landing Zones and resource hierarchy in Enterprise-Scale mode (Hub & Spoke)
- Implementation of collaborative tools (Azure DevOps, Git) to optimize project management
- Evaluation and integration of new Azure technological advances to maintain high-performance and sustainable solutions
- Documentation of processes, standards and procedures to ensure consistent and standardized management

Deployment and Automation with Infrastructure as Code (IaC)

- Configuration of Azure Pipelines and use of ARM templates (JSON) to automate infrastructure deployments
- Creation of Azure DevOps organization and implementation of Azure Repos repositories for storage and management of application and infrastructure scripts

- Implementation of Infrastructure as Code (IaC) for automatic resource deployment, including Management Group and Subscription
- Implementation of compliance strategies (Azure Policy) directly in ARM templates to manage access and secure resources
- Automation of repetitive processes via PowerShell and Azure CLI to minimize manual interventions

Security and Compliance

- Analysis and integration of Azure security services (Azure Key Vault, NSG) to meet information system security requirements
- Implementation of security policies to ensure resource compliance and assist teams in their integration

Data Management and Pipeline Optimization

- Design and optimization of data and BI pipelines to ensure continuity in feeding applications and visualization tools
- Use of Azure Data Factory and APIs to develop data integration solutions via ETL processes
- Data modeling and normalization to meet business team needs

Resource and Cost Optimization

- Evaluation of resource usage costs, virtual machine adjustment and choice of appropriate storage solutions
- Implementation of automatic management strategies to optimize costs by deactivating inactive resources and adjusting Power BI licenses according to needs
- Creation of dashboards for cost monitoring and budget forecasting

System Monitoring and Maintenance

- Exploitation of Azure monitoring tools (Monitor, Log Analytics) to identify and optimize overload points
- Configuration of performance alerts to anticipate and quickly resolve critical incidents
- Continuous supervision of Data & BI solutions to ensure their optimization and availability

Azure Environment and Services: Azure Active Directory (AAD), Azure AD Connect, Management Group, Subscription, Azure RBAC, Azure Policy, Azure Virtual Network (VNet), Azure Firewall, Route Table, Network Security Group (NSG), Azure Key Vault, Azure Monitor, Azure Log Analytics, Azure Storage, ARM Templates, Azure DevOps, Azure PowerShell, Azure Kubernetes Services, Azure Container Registry, Terraform, Azure Synapse, Azure Data Factory, Azure Data Lake Store Gen2, NoSQL, HDFS, Apache Spark, SQL Server, Apache Kafka, PyCharm, Python, REST/RESTful, Snowflake, Power BI

JANUARY 2023 – JUNE 2023 | CHARENTE MARITIME DEPARTMENT, FRANCE

Data Architect

Move2Cloud Project: Implementation of hybrid On-Premises/Azure platform and automation of infrastructure and application deployment with CI/CD strategy using Azure services

Target Architecture Design

- Participation in requirements gathering meetings and architecture definition using Microsoft Cloud Adoption Framework recommendations and best practices for Azure
- Definition of hierarchical structure hosting initial Azure Landing Zones
- Participation in defining initial compliance strategy for deployment at Management Group and Subscription Azure Policy levels (RDP Access, Azure SQL encryption, etc.)
- Definition of Azure services to deploy in landing zones in Enterprise-Scale target architecture with Hub & Spoke topology
- Study of various financial options made available by Microsoft publisher for Cloud platform cost optimization (Azure Hybrid Benefit, License Mobility, Reserved Instances, etc.)
- Definition of migration strategy by iteration of part of On-Premises applications
- Choice of GDM application (Workforce Management) as migration pilot project to deploy with Rehost pattern (in IaaS)

Azure Landing Zones Deployment with Azure DevOps

- Recovery of Billing Account identifier linked to Azure Enterprise Agreement (EA) account to automate Azure subscription creation
- Creation of Azure DevOps Organization, projects and Azure Repos repositories for source management (Platform and Application scripts)
- Implementation of Infrastructure as Code (IaC) using ARM Templates (Azure Resource Manager) and Azure PowerShell scripts (Az PowerShell Module) enabling automatic deployment of various services, hierarchical levels (Management Group and Subscription) and Azure landing zones
- Integration of initial compliance strategy in ARM Templates at Management Group and Subscription levels with Azure Policy
- Deployment of GDM application for migration pilot project on Azure
- Implementation of cost monitoring Dashboards to control budget and make future projections

Azure Environment and Services: Azure Active Directory (AAD), Azure AD Connect, Management Group, Subscription, Azure RBAC, Azure Policy, Azure Virtual Network (VNet), Azure Firewall, Route Table, Network Security Group (NSG), Azure Key Vault, Azure Monitor, Azure Log Analytics, Azure Storage, ARM Templates, Azure DevOps, Azure PowerShell, Azure Kubernetes Services, Azure Container Registry, Azure Synapse, Azure Data Factory, Azure Data Lake Store Gen2, NoSQL, HDFS, Apache Spark, SQL Server, Apache Kafka, PyCharm, Python, REST/RESTful, Snowflake, Power BI

JANUARY 2022 - DECEMBER 2022 | BOMBARDIER, MONTREAL, CANADA

Data Architect

Project: Azure Platform Administration, CI/CD Automation, Data Engineering, Data Modeling and DataOps

CI/CD Process Automation with Azure DevOps

- Implementation of CI/CD pipelines via Azure DevOps YAML pipelines for infrastructure deployment

DataOps & Power BI Administration for Modern Data Platform

- Design, build, test and deploy Data & BI solutions used for various business sectors and applications in ongoing and future projects
- Design, develop, deploy and optimize data and BI pipelines used to feed data applications or data visualizations
- Ensure execution and maintenance of Data & BI solutions and Data & BI products
- Design, develop and deploy visualization and analysis tools, as well as automation, to facilitate DevOps on Azure Data Platform
- Share best practices in Data & BI processing and data storage with other data engineers
- Design and develop data integration and consumption solutions using APIs, ETL jobs and/or streaming (ETL pipelines with Python / Spark / Azure Data Factory)
- Develop and maintain scalable data pipelines and create new integration APIs to support continuous growth in data sources, volume and complexity
- Model and normalize data to meet team and business line needs
- Diagnose problems and solve complex issues ensuring developed solutions are sustainable with a continuous improvement perspective
- Implement continuous deployment strategies while ensuring high availability of production systems
- Participate in architecture choices for implementing analytical projects under Power BI, including choice of Power BI license types (Free, PRO and Premium)
- Develop Power BI Dashboards exposing customer-related indicators
- Calculate and implement analysis and monitoring strategy for customer KPIs using tabular cubes and Power BI
- Administer Power BI service: pipelines, deployments, workspaces, security, Gateway, Premium Capacity

Azure Environment and Services: Snowflake, Azure VM, Azure VNet, PyCharm, Python, REST/RESTful, NoSQL, HDFS, Apache Spark, SQL Server, Apache Kafka, Agile, Scrum, Jira, Git, Azure DevOps, Linux, Windows, Docker, Azure: Azure Synapse, Azure Data Factory, Azure Data Lake Store Gen2, Azure Function, Databricks, Delta, Power BI

AUGUST 2021 - DECEMBER 2021 | VIDEOTRON, MONTREAL, CANADA

Big Data Architect

Project: Data Platform Implementation

Data Engineering & Analytics

- Participate in defining data and BI solutions for ongoing and future projects
- Design, build, test and deploy Data & BI solutions used for various business sectors and applications in ongoing and future projects
- Design, develop, deploy and optimize data and BI pipelines used to feed data applications or data visualizations
- Ensure execution and maintenance of Data & BI solutions and Data & BI products
- Share best practices in Data & BI processing and data storage with other data engineers
- Design and develop data integration and consumption solutions using APIs, ETL Jobs and/or Streaming (e.g., ETL pipelines with Python/Spark/Azure Data Factory)
- Develop and maintain scalable data pipelines and create new integration APIs to support continuous growth in data sources, volume and complexity
- Diagnose problems and solve complex issues ensuring developed solutions are sustainable with a continuous improvement perspective
- Implement continuous deployment strategies while ensuring high availability of production systems
- Administer Power BI service including pipelines, deployments, workspaces, security, Gateways, Premium capacity, etc.

Azure Environment and Services: Azure VM, Azure VNet, Azure Key Vault, Azure Kubernetes Service, Docker, Git, Bitbucket, PyCharm, Python, REST/RESTful, NoSQL, HDFS, Apache Spark, Apache Zookeeper, SQL Server, SAS, Apache Kafka, Apache Airflow, Agile, Scrum, Jira, Linux, Windows, Hortonworks: Apache Ambari, HDFS, Hive, Oozie, Yarn, Kerberos, Ranger, Zeppelin, Amazon S3, Spark MLlib, Scikit-learn, TensorFlow, NumPy, Pandas

JUNE 2018 - JULY 2021 | CBC-RADIO CANADA, MONTREAL, CANADA

Data Architect

Project: CBC-Radio-CANADA Azure Architecture Evolution & Data Platform Implementation

Azure Architecture Evolution

- Management of Azure contract and subscriptions with Microsoft
- Propose FinOps Best Practices to optimize Azure platform costs (Dev/Test subscription, scheduled shutdown of Azure services on non-production environments, reuse of existing licenses, etc.)

- Refactoring of IaaS applications to optimize their architecture
- PoC for modernization of monolithic application with development team for containerized Microservices deployment on Azure Kubernetes Service (AKS) cluster
- Support teams in implementing DevOps culture and tools for automated deployments on Azure

Data Engineering and MLOps on Azure & GCP

- Implementation and administration of data lake in Microsoft Azure Data Lake Store
- Design and development of data integration and consumption solutions using APIs, ETL Jobs and/or Streaming (ETL pipelines (Python / Spark, Azure Data Factory))
- Classification of digital documents (press articles) using natural language processing (NLP) techniques
- Design and implementation of continuous integration and continuous development DevOps patterns ensuring integration with other improvement tools in software development lifecycle
- Data modeling and normalization to meet team needs
- Problem diagnosis and complex issue resolution ensuring developed solutions are permanent with continuous improvement perspective
- Implementation of continuous deployment strategies while ensuring high availability of production systems
- Contribution to automation of continuous integration, automated testing and continuous deployment processes

Azure Environment and Services: Azure VM, Azure VNet, Azure Key Vault, Azure Kubernetes Service, Docker, Git, GitHub, IntelliJ, Java, Python, Scala, REST/RESTful, Maven, NoSQL, HDFS, Apache Spark, Apache Airflow, Apache Kafka, Apache Zookeeper, Apache Cassandra, ElasticSearch, GCP, Google BigQuery, SQL Server, Redis, MongoDB, Facebook Ads API, Agile, Scrum, Jira, Linux, Windows, Docker, Cloudera: Cloudera Director and Cloudera Manager, Cloudera Navigator, Hadoop, Hive, Impala, Sqoop, Oozie, Yarn, Kerberos, Sentry, Power BI, Spark MLlib, OpenNLP, NLTK, Scikit-learn, Gensim, Google Natural Language API

NOVEMBER 2017 - MAY 2018 | QUEBECOR MEDIA, MONTREAL, CANADA

Big Data & ML Consultant

Project: Implementation of Content Recommendation and Personalization Platform

Big Data & ML Platform Implementation

- Implementation of content recommendation and personalization platform
- Develop and implement real-time data processing algorithms
- Implementation of learning algorithms: Machine Learning
- Automatic natural language processing using Google NLP API

- Data analysis and Data mining
- Data modeling, exploration, and preparation
- Information retrieval and personalized information retrieval (Behavioral Search)

Environment: IntelliJ, Java (Spring), Python, Scala, REST/RESTful, NoSQL, HDFS, Apache Spark, Apache Kafka, RabbitMQ, Apache HBase, Apache Zookeeper, Apache Cassandra, ElasticSearch, Google BigQuery API, SQL Server, Redis, MongoDB, Agile, Scrum, Jira, Git, Linux, Windows, Docker, Microsoft Azure, Google Cloud Platform, Amazon Web Services, Apache PredictionIO, Spark MLlib, OpenNLP, NLTK, Scikit-learn, Gensim, Google Natural Language API

JULY 2014 - OCTOBER 2017 | CEDROM-SNI, MONTREAL, CANADA

R&D Developer in Big Data

Big Data Platform Implementation for R&D Department

- Development and implementation of real-time data processing algorithms
- Implementation of learning algorithms: Machine Learning
- Automatic natural language processing (NLP)
- Data analysis and Data mining
- Creation and evolution of data analysis algorithms on "Big Data" type infrastructures fed by supervision and audit data
- Data indexing and search on ElasticSearch
- Improvement and optimization of document search on web portal (http://eureka.cc)
- Participation in collection, processing, and provision of more than 2 million documents (Paper press, Blogs & Websites, Social networks, TV & Radio, Reports...) daily
- Development, deployment and continuous improvement of document capture, enrichment solutions, and their provision in our databases or search engine
- Production implementation and management of ElasticSearch cluster
- Recovery of evolution requests, specify, implement, test and support production implementation of developed features (technologies: HDFS, Hadoop, HBase, Zookeeper, and Apache Storm/Apache Kafka)
- Design and development of programs in Python / XML / XSLT
- Implementation of website "Crawling" platform
- Implementation of Microservices architecture with Docker
- Administration of Microservices architecture and autoscaling with Apache Mesos, Mesos-DNS and Marathon
- Data modeling, exploration, and preparation

- Information retrieval and personalized information retrieval (Behavioral Search)

Environment: Eclipse, Java, REST/RESTful, Python, Django framework, ElasticSearch, Logstash, Kibana, Redis, Apache Kafka, Scala, Apache Storm, Spark, Apache Zookeeper, SQL Server, Agile, Git, Jira, Linux, Amazon Web Services, Windows, Apache Mesos, Mesos-DNS, Marathon, OpenNLP, Stanford CoreNLP, Spark MLlib, NLTK, OpenCV, Scikit-learn, Gensim

JANUARY 2013 - APRIL 2014 | MORNEAU SHEPELL, MONTREAL, CANADA

C#/VB .NET Developer

System and Application Development

- Familiarize with existing program code
- Program according to previously performed analysis considering current programming standards
- Verify results to ensure proper program functioning
- Participate in documentation writing
- Use, manipulate and parse XML files (messages, results... (Linq to XML))
- Develop HTML forms
- Create unit and integration tests
- Meet user needs

Environment: Visual Studio 2010, C#, WCF, VB.net (.NET 4), MS CRM, WebServices, REST, SOAP JavaScript, SQL Server, Agile, Windows

MAY 2011 - OCTOBER 2012 | CEDRIC LABORATORY, PARIS, FRANCE

Research Assistant

Participation in R&D Activity of Research Direction

- Word clouds and digital libraries
- Write state of the art of subject and specifications
- Develop scripts to clean raw data
- Extract data
- Develop application for interactive data visualization
- Evaluate results and tests

Environment: Eclipse, Java, Python, Processing, Linux, TreeTagger

FEBRUARY 2008 - MAY 2008 | INDUSTRY PROMOTION AGENCY, TUNIS, TUNISIA

PHP/JavaScript/MySQL Developer

- Implementation of intranet for business incubator activity management
- Analyze needs and write specifications
- Write technical and functional specifications
- Do UML design with Rational Rose
- Class diagrams, Sequence diagrams and use cases
- Develop in PHP, HTML, XHTML, CSS, Javascript
- Develop MySQL database and Unit tests

Environment: PHP, HTML, CSS, JavaScript, MySQL, Windows, UML, Rational Rose

OCTOBER 2006 - APRIL 2007 | SKILLS INSTITUTE FOR CONSULTING AND TRAINING, TUNIS, TUNISIA

VB Developer

- Development of vehicle fleet management application in VB 6
- Needs analysis and specification writing
- Technical and functional specification writing
- Design (MERISE) and development of SQL Server 2005 database
- Unit tests

Environment: VB6, SQL Server 2005, MERISE

OCTOBER 2005 - APRIL 2006 | SKILLS INSTITUTE FOR CONSULTING AND TRAINING, TUNIS, TUNISIA

Intern

- Initialization and exposure to Java development environment

Environment: Java, Eclipse

TECHNICAL SKILLS

Cloud, Data & ML

Azure Active Directory (AAD), Management Group, Subscription, Azure RBAC, Azure Policy, Azure Virtual Network (VNet), Azure Firewall, Route Table, Network Security Group (NSG), Azure Key Vault, Azure Monitor, Azure Log Analytics, Azure Storage, ARM Templates, Azure PowerShell, Azure Kubernetes Services, Azure Container Registry, Terraform, Databricks, Snowflake, Azure Data Lake Storage Gen1/Gen2, Azure Data Factory, Hortonworks, Cloudera Data Platform, Amazon Web Services, Amazon S3, Apache Spark, Apache Kafka, RabbitMQ, Apache HBase, Apache Zookeeper, SQL Server, Apache

Cassandra, ElasticSearch, Logstash, Kibana, Google BigQuery API, SQL Server, Redis, MongoDB, Apache PredictionIO, Spark MLlib, OpenNLP, NLTK, Scikit-Learn, Gensim, Pandas, Google Natural Language API

DevOps & Orchestration

Git, GitHub, Team Foundation Server, Azure DevOps, Docker, Kubernetes (AKS)

Programming Languages

SQL, C#, VBA, Python, Java, R

Methodology

Agile SCRUM, Atlassian Confluence, Atlassian JIRA

EDUCATION AND CERTIFICATIONS

2016 | Graduate Diploma in Information Technology
École de Technologie Supérieure de Montréal (ETS) - Montreal - Canada

2011 | Master 2 in Computer Science - Modeling, Optimization and System Design
CNAM Paris - University of Versailles-Saint-Quentin-en-Yvelines (UVSQ) - France

2008 | Bachelor's Degree in Computer Science Applied to Management
École Supérieure des Sciences Économiques et Commerciales (ESSEC) Tunis Tunisia

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LANGUAGES

ENGLISH: Read, Written, Spoken